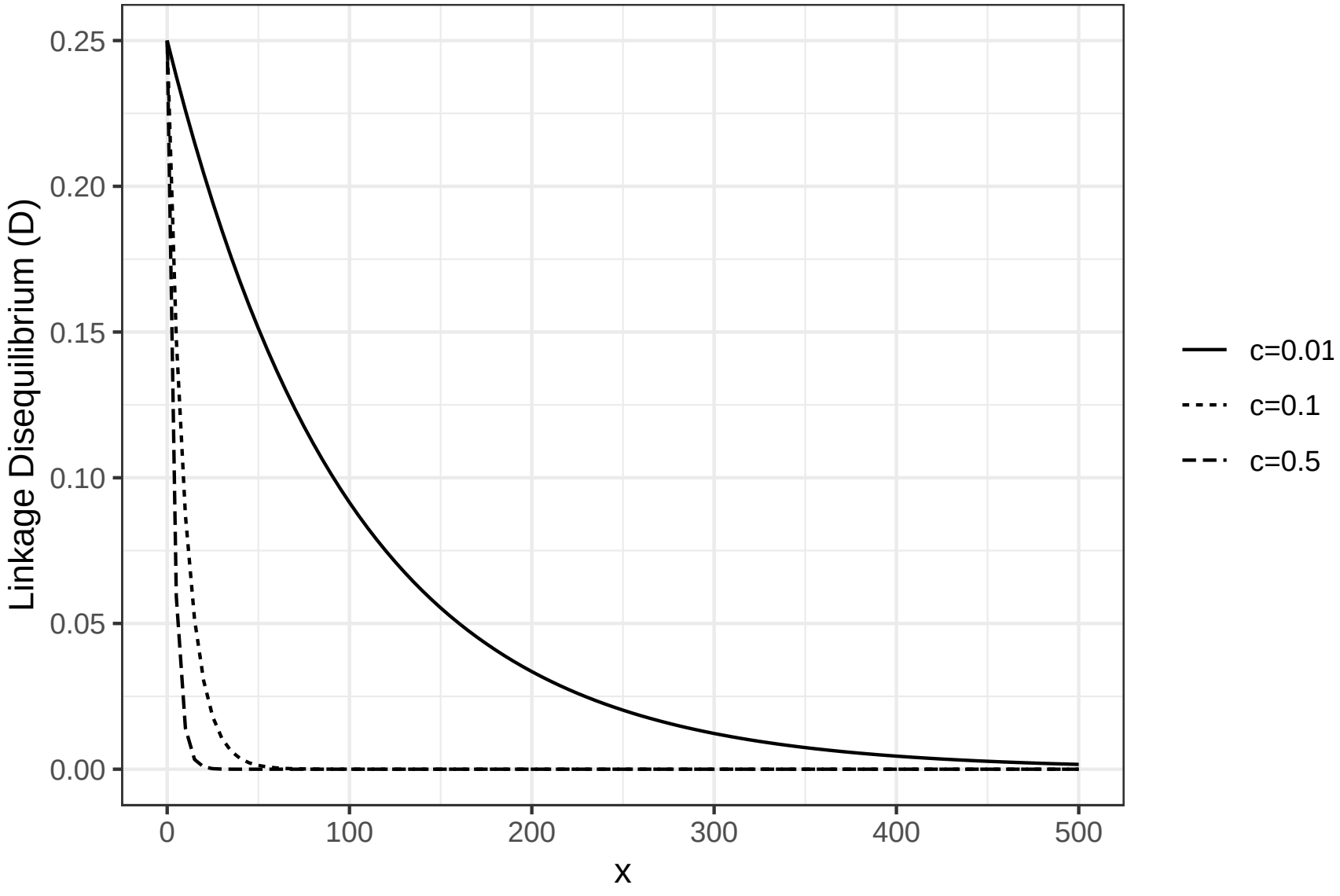
LD Decay through time



Linkage Disequilibrium Simulator

link: https://cjbattey.shinyapps.io/LDsim/

Linkage and Linkage Disequilibrium

new population | next generation

recombination rate

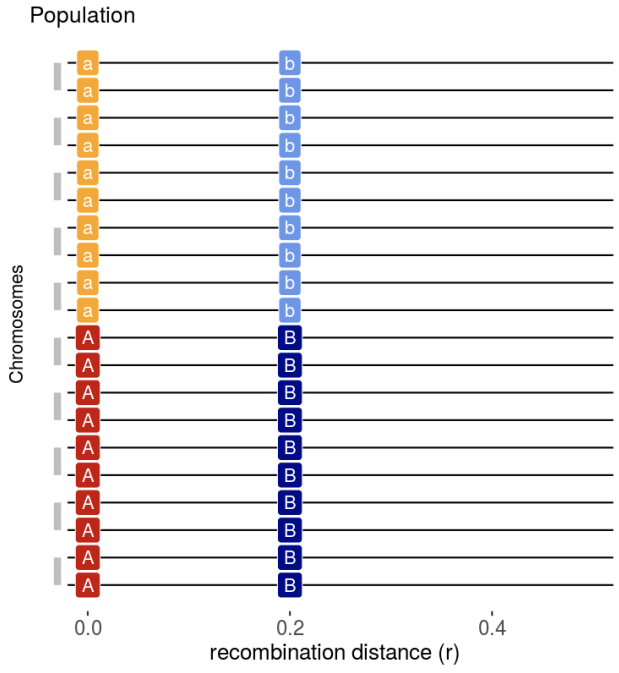
0 | 0.2 | 0.5

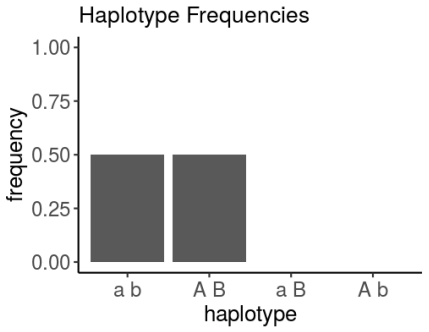
0 0.05 0.1 0.15 0.2 0.25 0.3 0.35 0.4 0.45 0.5

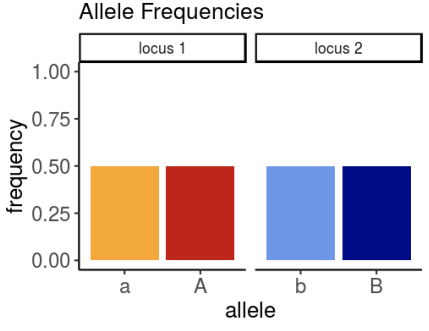
chromosomes | frequency tables | Change Over Time

Here's a diploid population where we're tracking two loci (regions of the genome). At each locus there are two alleles -- A/a and B/b. In the first generation all of the chromosomes are either AB or ab. Recombination generates new combinations of alleles (haplotypes) by swapping pieces of chromosomes during meiosis -- shown here with crossing purple lines.

Try clicking 'next generation' a few times and see what happens to the population. Does the number of haplotypes generally go up or down? What happens when we change the recombination rate (r)?







LDSim Activity

1) Under the "Chromosomes" tab, look at the plot that appears when the website loads. What is the x axis? What does each horizontal line represent? What do the gray bars pairing horizontal lines represent? How about the colors?

2) Using the same "Chromosomes" tab, move the "recombination rate" slider to 0, then click "Next Generation" several times. What happens to haplotype and allele frequencies? Click "New Population" to reset, and change the recombination rate to a higher value and repeat the simulation. How does your answer change?

3) Navigate to the "Change Over Time" tab and click "Run". What does the graph that appears below the sliders show? Describe all features.

4) Under the "Change Over Time" tab, run simulations using different combinations of recombination rate and population size. Which values lead to the quickest loss (or "decay") in LD? Which values lead to the slowest?

5) Develop a question that can be answered using this app.